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Declaration

Acknowledgements

Chapter 1

Abstract

Chapter 2

Gravitational Wave Interferometers

This chapters purpose is to link general relativity to a basic michelson interferometer. I will keep the general relativity to a minimum, since this isn't a textbook on GR, however there is a small amount required to know why gravitational wave detectors are so big, why they are L shaped, etc. It will then expand on the basic michelson by adding the cavities. This is a good point to link sources to the detector, since the signal recycling adds shaping, its good to know why we aimed for 100Hz (I actually don't know why this was done, for some reason LIGO is optimised for detecting heavy black holes I think). Talk about noise sources, noise and sensitivity curves, transfer functions between differential arm motion to light modulation at the AS port, mention the standard quantum limit. A detailed explanation of shot noise and radiation pressure noise should be in here, since these basically give you the requirements for the laser chapter and the photodiode chapter. Once the noise sources are covered, the BNS range can be shown. It makes sense to introduce the current detectors in more detail, and then talk about the 3rd generation of detectors. More novel topologies can be mentioned after.

2.1 Basic Michelson interferometer as GW detector

2.1.1 dark fringe

2.2 Michelson with arm cavities

2.3 Michelson with power recycling

2.4 Michelson with signal recycling

2.5 LIGO

2.6 Sensitivity curves

2.7 Noise sources

2.8 Third Generation Ground based Detectors

2.9 other detectors

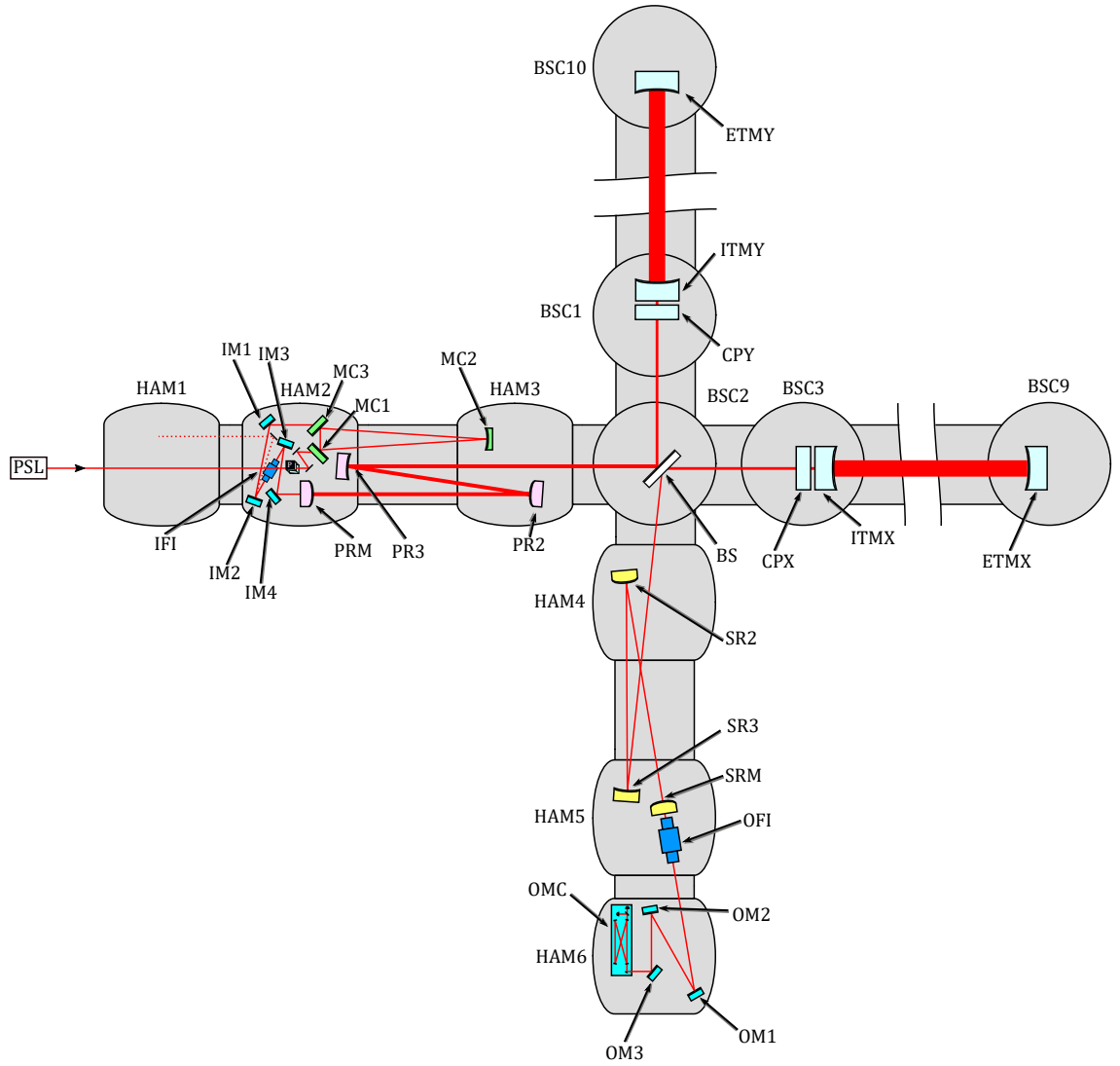


Figure 2.1

Chapter 3

A Prototype High Power Laser for LIGO's Fourth Observation Run

3.1 Introduction

The LIGO detectors require high input power, nominally 125 W, at 1064 nm to decrease the shot noise which limits its sensitivity at frequencies above 100 Hz. If a laser is in a coherent state, the shot noise associated with detecting the laser is $\sigma = \sqrt{\frac{hc}{\lambda}P}$, so the signal to noise ratio at high frequencies scales with $\sqrt{P}$. Alternatively, to decrease the shot noise limit the wavelength could be decreased; however, the LIGO detectors operate at 1064 nm due to the availability of mirror coatings with low absorption and low thermal noise. In fact, future interferometers may operate at longer wavelengths due to the potential reduction of mirror coating thermal noise that can be achieved at longer wavelengths [1].

In the field of gravitational wave detection, high power refers to lasers with powers of the order 100 W. High power is a loosely defined term as lasers ranging from 100 W to 100 kW are defined as high power in the literature [2–12]. As the power is increased, other properties of the laser tend to be degraded. It needs to operate uninterrupted on year-long timescales; this poses a major engineering challenge.

The laser for LIGO must be continuous wave, linearly polarised, have high spatial purity, low intensity noise and low frequency noise. These requirements are detailed in [13]. To overcome the technical challenges that come with the high power and low noise requirements for the LIGO laser, the master-oscillator-power-amplifier (MOPA) topology was selected. This type of laser is based on a high performance seed laser which is then amplified. As amplifiers are linear and coherent, the low noise properties of the seed laser characterises the noise of the high power beam.

LIGO’s original laser used a high power oscillator (HPO) as the means for obtaining the target power. The HPO was a ring cavity containing laser crystals that was seeded by a 35 W low noise laser. This design went through extensive testing and refinement at the AEI [14] and was installed as part of LIGO’s pre-stabilised-laser (PSL). However, due to jitter noise from the water cooling of the crystals within the HPO coupling to the differential arm motion signal (DARM), as well as one HPO being destroyed by accident, the HPO was bypassed and the LIGO detectors operated using the 35 W laser [15].

To reach higher powers during LIGO’s third observation run, instead of the HPO, a neoVAN-4S single-pass amplifier was used. This stepped up the laser power from 35 W to 70 W [16]. The next stage of development was a complete redesign of the

LIGO laser based on using two upgraded versions of the neoVAN-4S. The neoVAN-4S-HP amplifier showed promising results [11, 12], so a prototype laser was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of LIGO’s fourth observation run [15].

In this chapter the new laser layout is discussed. The PMC throughput power, beam quality, intensity noise, pointing noise and frequency noise of the prototype laser were measured. It was found that this layout provided a suitably low noise laser. This laser was able to produce 100 W of intensity stabilised light. While the 125 W nominal power level was not obtained, additional work in optimising the layout may improve the maximum output power of this laser.

3.2 Concept for LIGO’s New Laser

LIGO will use a MOPA laser that is seeded by a non-planar ring oscillator (NPRO) laser. The NPRO beam will be amplified by two single-pass amplifiers in series and be followed by a mode cleaner cavity to produce a low-noise high-power laser in the fundamental mode. This concept is shown in Figure 3.1. As the NPRO has good free running noise characteristics the amplified beam will as well. Results from a setup using a single neoVAN-4S-HP showed these amplifiers can be continuously operated for many tens of days [12]. The lead to the proposal to construct the PSL out of two of these in series. A similar set up using a 35 W seed laser and two neoVAN-4S-HPs was able to generate 195 W of light [11] for nearly 80 days with no significant drops in power.

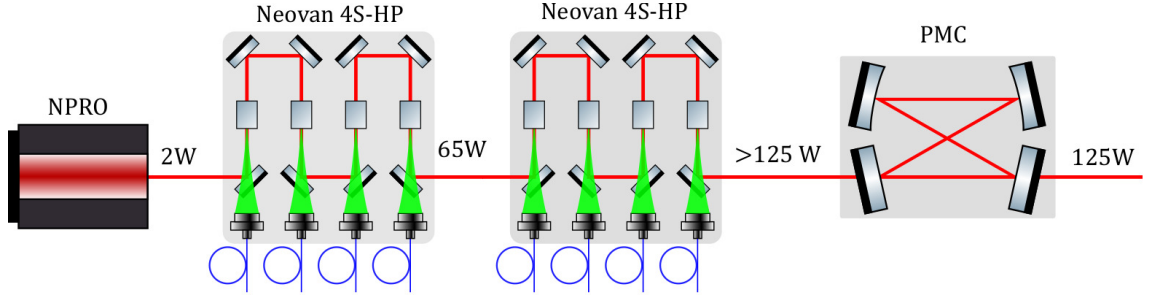


Figure 3.1: Conceptually, the PSL will be made up of four components: the NRPO, two neoVAN-4S-HP amplifiers and the PMC cavity. The NPRO provides seed light with good noise characteristics. These noise characteristics are preserved by the amplifiers, so the resulting beam will have high power and low noise. The PMC filters junk light and RF intensity noise so a spatially pure mode can be set to the interferometer.

3.2.1 Non-Planar Ring Oscillator Lasers

The first element of the PSL is a non-planar ring oscillator laser [17–19]. The NPRO provides a low noise laser making it a suitable seed for a MOPA system. This is a single frequency laser constructed from one Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with high amounts of frequency stability. The NPRO used was a Mephisto, built by Coherent, and it can produce to 2W of 1064 nm light.

3.2.2 The neoVAN-4S-HP Solid-State Amplifier

To amplify the NPRO beam, two neoVAN-4S-HP amplifiers will be used. The neoVAN-4S-HP is a single-pass, solid-state amplifier manufactured by Neolase. It is constructed from four Nd : YVO₄ crystals which are pumped by fibre coupled,

low noise laser diodes. It should be seeded by s-polarised light at 1064 nm. These gain mediums are in contact with a metal plate which is cooled. This keeps jitter noise low as flowing water is not directly in contact with the gain medium. As these amplifiers have low noise pump diodes and will be seeded by an NPRO, the noise of the light produced by them should be determined by the NPRO.

3.2.3 The Pre Mode Cleaner Cavity

The pre mode cleaner (PMC), see Figure 3.2, is used as the hand-off point between the PSL and the rest of the interferometer. Light transmitted by this cavity is used as the input beam for the interferometer. The PMC is a bow tie cavity with an FSR of 9 MHz, thus it passively filters RF intensity noise. Control side bands are provided after the PMC but before the IMC, meaning that the IMC cannot be used to provide this stabilisation. The PMC is locked to the fundamental mode so the beam that is delivered to the rest of the interferometer has a low amount of higher order mode content. This is important as the amplifiers produce light with spatial defects. The PMC has been redesigned [20] to prevent contamination due to glue on its mirrors and to make maintaining the PMC easier. The PMC design no longer includes a vacuum enclosure.

3.3 Layout of the Prototype Laser System

A sketch of the layout is shown in Figure 3.2. In addition to the main beam path, there are low power diagnostic beams after each amplifier. The prototype layout did not include the tabletop frequency stabilisation servo (TTFSS) so this has not been included in the layout description. However, there is space for the TTFSS on

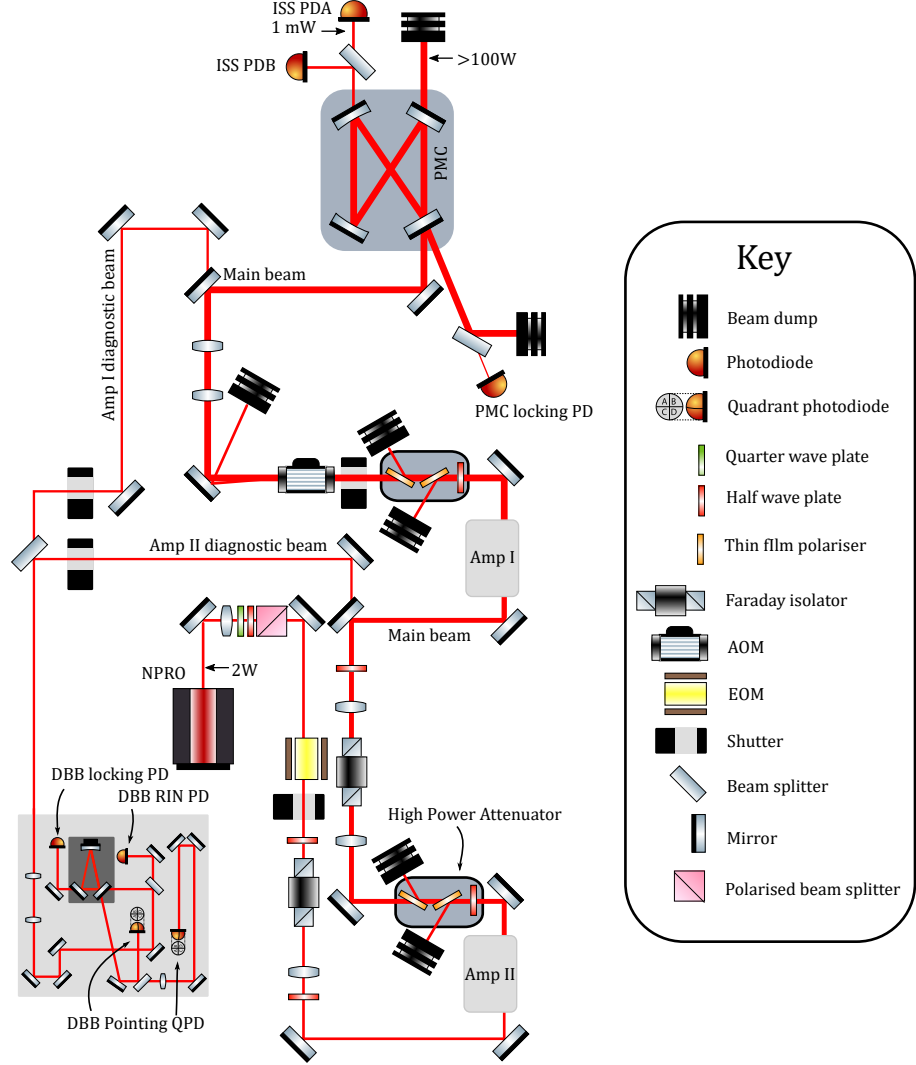


Figure 3.2: Sketch of the prototype optical layout that was constructed in the TnT lab. The main beam path and diagnostic beam paths are shown. Key photodiodes are also shown. A to-scale version of this layout is found in [21].

the optical table.

The NPRO beam passes through some optics to clean up the polarisation. Transmission through a steering mirror is used to monitor the power of the NPRO. An EOM is used to produce 35.5 MHz sidebands for the Pound-Drever-Hall (PDH) scheme used to lock the PMC. A Faraday isolator ensures light reflected back from the amplifiers does not damage the NPRO. The beam passes through lenses to mode match the beam to the first amplifier. A half-wave plate ensures the polarisation into the first amplifier is set correctly. These optics are shown in figure 3.3.

After the first amplifier, the beam passes through a high power attenuation stage allowing for safe set up and low power operation. When setting up this layout, the amplifiers must be run at the maximum power since they exhibit significant thermal lensing. The high power attenuators are not envisioned being used during observation runs. The laser then is directed through another Faraday isolator and wave-plate before reaching the second amplifier. The mode matching lenses for the second amplifier are on either side of the Faraday isolator. The transmission through one of the steering mirrors before the second amplifier is used for monitoring the beam produced by first amplifier. This section of the layout shown in Figure 3.3.

After the amplifiers, the beam goes through another shutter and high power attenuation stage before passing through an AOM. This AOM is used for modulating the power of the laser. After the AOM, a pick-off is used to measure the properties of the beam just before it goes into the PMC. This part of the the layout can be seen in figure 3.4.

The beam is coupled to the PMC cavity. There are three transmission beams from the PMC, a high power one and two low power ones. One of the low power beams is used by inner intensity-stabilisation-servo (ISS). The other would be used by the TTFSS. The PMC can be seen in figure 3.5. The reflected light is used to lock the PMC, this path can be seen in figure 3.4.

Diagnostic beams are picked off from the main beam. The diagnostic bread board (DBB) is able to make intensity noise, frequency noise, beam pointing noise and higher order mode content measurement [22, 23]. The DBB is shown in figure 3.6. The breadboard consists of a cavity, motorised lenses, an automatic alignment system and several photodiodes. Despite having motorised lens mounts in the DBB, static mode matching lenses are required in the diagnostic beam paths. It is integrated into the PSL via several pick-off points and can be used to monitor the status of the NPRO, Amp I and Amp II.

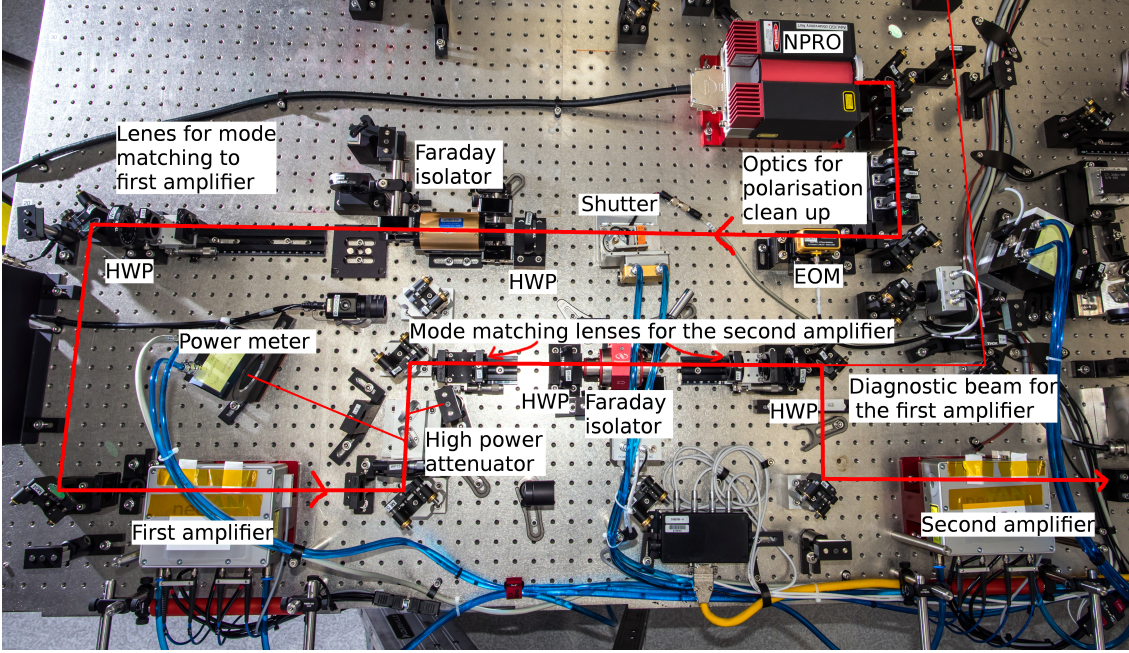


Figure 3.3: Photograph of the PSL prototype showing the beam path from the NPRO to the second amplifier. The beam produced by the NPRO passes through some optics that clean up the beam's polarisation.

Figure 3.4: Photograph of the PSL prototype showing the beam path from the second amplifier (Amp I) to the input of the PMC. The path taken by reflected light from PMC used to lock the PMC can also be seen.

Figure 3.5: Photograph of the PMC and the inner ISS photodiode box.

Figure 3.6: Photograph of the DBB and beam paths towards it.

3.4 The PMC Output Power Reached by the Prototype Layout

The PSL should deliver 125 W to the interferometer [13]. There are several variables that will affect the amount of power that is transmitted through the PMC: the flow rate through the amplifiers heads; the input alignment to each amplifier; the mode matching to each amplifier; the mode matching to the PMC; the alignment to the PMC; the pump diodes current and the polarisation of the light that is to be amplified. These are not all independent of each other. For instance, changing the flow rates will change the thermal lensing in the amplifier. This will change the optimal mode matching and alignment to the amplifier. Changing the the pump diode current effects the thermal lensing as well, meaning increasing it doesn't always lead to an improvement in TM00 power.

The prototype layout was able to deliver a maximum of 111 W through the PMC during the a 55 hour trial run. This is shown in Figure 3.7. The power was measured with recently calibrated power meters. The power dropped to 107 W through the PMC. Changes in the lab temperature and humidity caused slight alignment changes. This resulted in the power decreasing. As the PSL is in a controlled environment, this effect will be less severe when commissioned for O4. The light produced by the amplifiers was greater than 125 W. Mode matching between the final amplifier and the PMC could be improved at the percent level. This power could be increased by increasing the flow rate and pump diode currents to the amplifier, however concerns of jitter noise meant that the flow rate was kept low. With higher flow rates, an input power of 137W with a higher mode content of 4.6% was

produced (Labbook post). To actuate on the lasers power, some of the incident light will need to be 'dumped'. This means that the full power will be reduced by ~ 10 W when the intensity stabilisation is implemented. Longer term measurements should be taken, however the results shown in [11] are promising. LIGO Hanford measured the input laser power dropping from 55 W to 52 W over the course of one year (cite Ninas new paper); this is likely due to mode mismatches as a result of changes in the set-ups thermal state/.

3.4.1 The Effect of Cooling on the Amplifier's Power and Beam Quality

Water flow example. The first thing that should be decided is how the cooling will be done and what flow rates will be used as the temperature of the amplifier heads will affect alignment and mode matching that is optimal. The flow rates of the coolant to the amplifier affects the the amount of thermal lensing. A balance must be struck between the amount of cooling - more is better - and the amount of jitter that can be potentially introduced to the beam via coolant moving through pipes. Due to the jitter noise experience at the Hanford detector that was due to the water cooling in the PSL, the flow rates will be set as low as possible. The thermal lensing in the Nd : YVO₄ is non-circular since they are cooled from below there is a significant temperature gradient across the crystals. For small amounts of thermal lensing, the beam will become slightly astigmatic. From a beam quality point of view, it is better to have more cooling as with more cooling there is less thermal lensing.

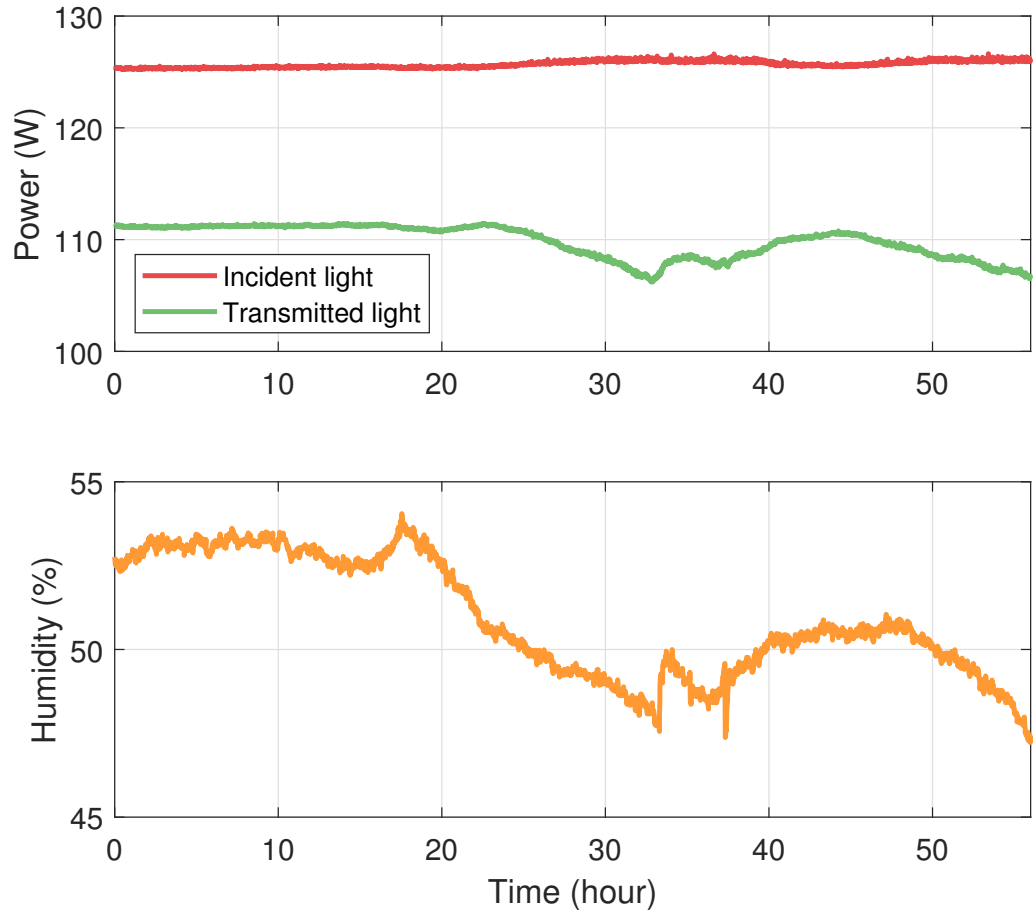


Figure 3.7: The top panel shows the power that was incident on the PMC and the power transmitted by the PMC over a 55 hour run. The light produced by the amplifiers did not change significantly for the duration of this measurement, however the transmitted light through the PMC dropped from 111 W to, at its lowest, 107 W. This is likely due to the environmental conditions in the lab changing. The bottom panel shows the humidity of the lab. Changes in the humidity, which can be caused by a temperature change, coincides with changes in the power level of the amplifier.

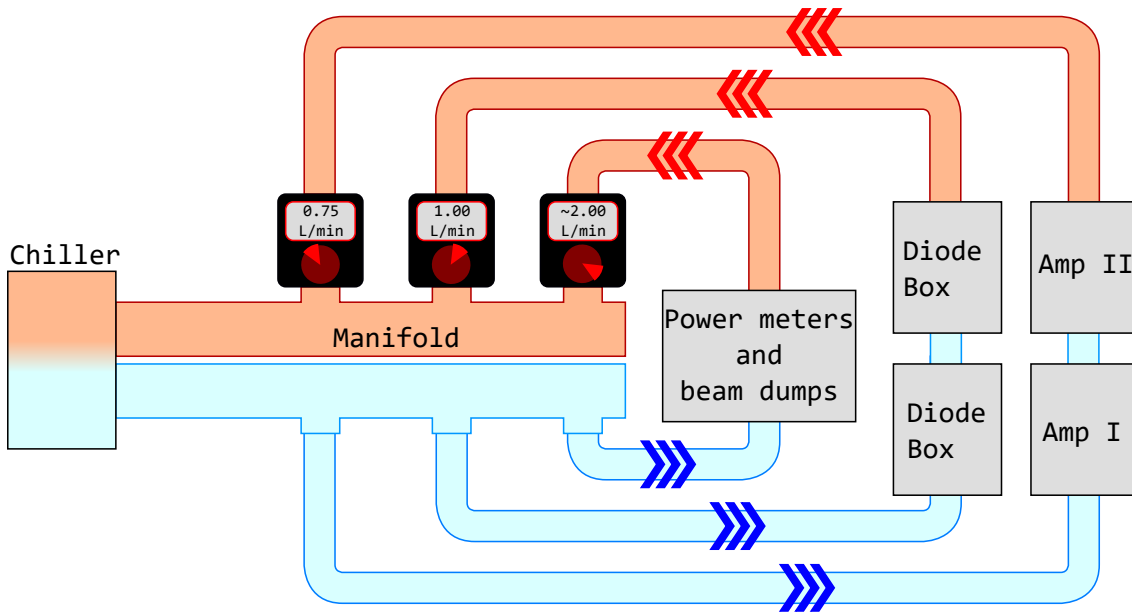


Figure 3.8: Cartoon of the cooling layout for the PSL. The amplifiers are on a separate loop to the diode boxes so that the flow rates for these can be reduced as much as possible. By placing the flow rate sensors on the return side of the the cooling system, leaks can be detected.

The cooling scheme that will be used is shown in Figure 3.8. Flow meters which monitor the flow rate of the coolant will be placed on the return side of the system rather than the supply side. By having the flow meters on the return side, they will measure a drop in flow if there is a leak, allowing for the water supply to be cut.

A different cooling layout was used to test what the appropriate flow rates for the amplifiers and diode boxes could be. The diode box and amplifier head were connected in parallel, and each amplifier was on a separate cooling loop. This allows for the flow rate through each amplifier independently, meaning that the effect of flow rate on the beam quality can be studied for one amplifier. labbook post

The flow rate to the first amplifier was kept high to get as high a quality beam as possible entering the second amplifier. The HOM content for this beam was 3.8%. The second amplifier and diode box flow rate was decreased. The temperature of the diode head was monitored. The higher order mode content and was measured with the DBB and the power was measured with a high-power power meter.

The change in beam quality and output power as a function of amplifier temperature is shown in Figure 3.9. As the flow rate was decreased, the temperature of the heads increased. At 22.3°C the higher order mode content was $\sim 4.5\%$ and the amount of power in the fundamental mode was $\sim 127\text{ W}$. By making mode scans, the change in higher order mode content was measured. For each amplifier head temperature, sixty mode scans were averaged. This revealed that changes in the second order mode made up the majority of the HOM content change. This is due to the changing in thermal lens focal length and the amount of thermally driven astigmatism of the beam. It was decided that to keep the amount of power in the fundamental mode high, the diode heads would be operated such that their temperature did not rise above $\sim 27^\circ\text{C}$.

It was decided that the flow rate for the amplifier heads would be set to 0.75 L min^{-1} , as this corresponds to less than 5% HOM and $\sim 125\text{ W}$ of power in the TM00 mode. If a lower flow rate was to be used, the temperature of the amplifiers heads would increase such that the power in the TM00 would fall below 125W. The flow rate to the diode boxes was set to 1 L min^{-1} ; this is as low as possible. If it is much lower then there is a run away heating up effect.

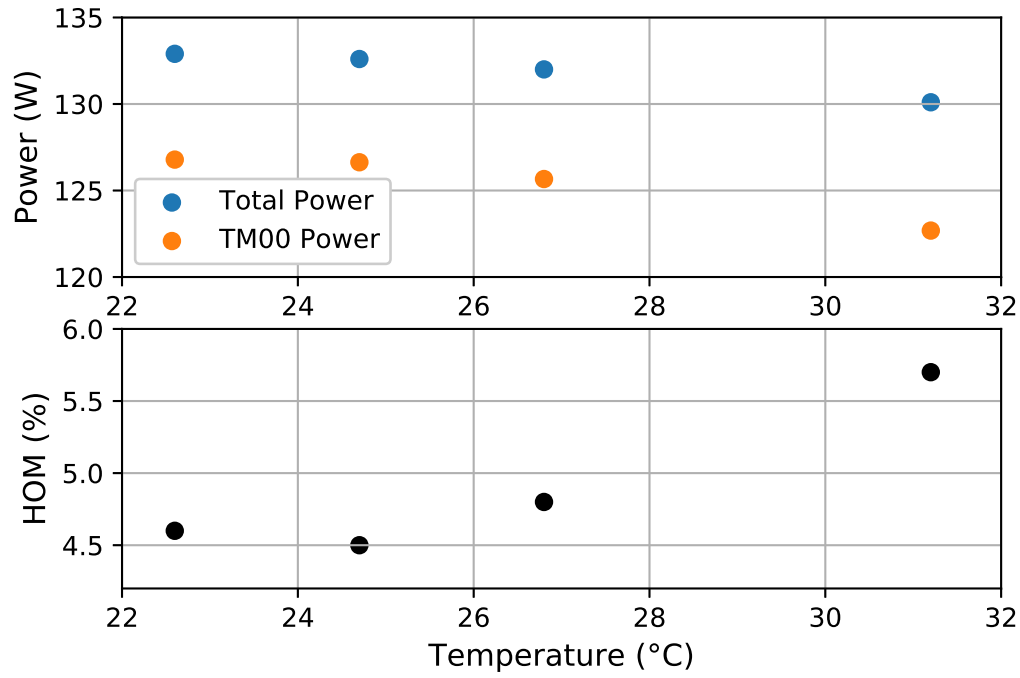


Figure 3.9: The top panel shows that as the second amplifier head's temperature increases the power produced by it decreases. The bottom panel shows that the higher order mode content increases as the temperature of the amplifier increases. This means that the power in the fundamental mode decreases as the temperature decreases due to two effects: less overall power and worse beam quality.

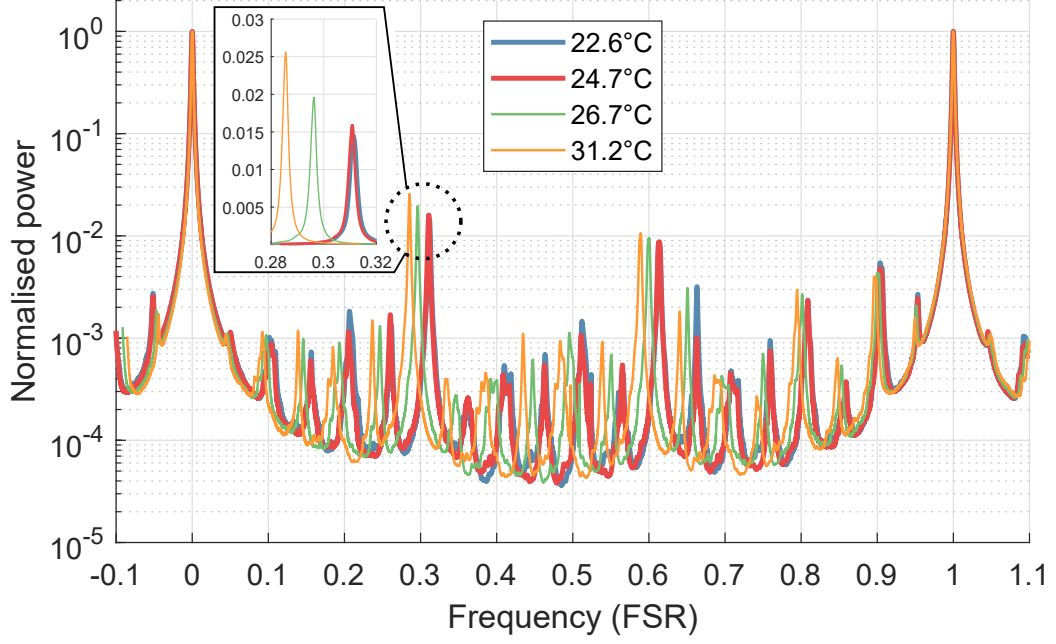


Figure 3.10: Mode scans were performed with the DBB for each amplifier head temperature. The second order mode for this cavity is located at ~ 0.3 FSR. As the temperature of the amplifier head changes, the second order mode content of the beam changes on the percent level. This second order mode height is shown in the inset. The second order mode can change due changes in the thermal lens focal length or because the beams astigmatism changes.

3.4.2 Amplifier Alignment Procedure

To obtain the high beam power low higher order mode content the alignment of the input beam to the amplifier needs to be optimised. Aligning amplifiers differs to aligning cavities as an amplifier does not have an eigenmode. The output power of the amplifier may increase at the expense of higher order mode content, so the beam profile and the beam power must be monitored when aligning the amplifier.

First, the amplifier should be switched off. This protects it against damage as well as preventing the possibility of unknown beams with several tens of Watts of power being in the lab. The beam path between the NPRO and the exit of the first amplifier were inspected to make sure the beam path was clear and terminated with dumps.

Next, the NPRO was set to produce 1.3W of light. This power of the light was measured on the input side of the amplifier. Using M03 and M04, the beam was roughly aligned so that the spot would hit the centre of the amplifier's input aperture. The power meter was placed on the exit side of the amplifier, and the input alignment was adjusted until the light exiting the amplifier was the same as the input power. Once this is done, one knows that the beam will be roughly aligned and some amplification will occur when the pump diodes are turned on. Once this was done, the high power attenuator immediately after the amplifier was set to dump all the light. The initial mode matching was based on the manufactures specifications, and the input light was set to s-polarisation.

The pump diode currents were increased until the amplifier ‘switched on’. This was at 2 A of current through each of the pump diodes. The amplifier starts amplifying the input light when there is sufficient pump light to cause enough population inversion that stimulated emission occurs.

At this threshold, the output power was maximised by optimising the alignment. Then, the mode-matching was adjusted, in tandem with correcting for misalignments caused by non-perfect centring of the beam on the mode matching lens, L02, and non-perfect co-alignment of the rail this lens is mounted on to the optical axis.

A smaller, faster power meter was used to monitor the power transmitted through the high power attenuator. Care must be taken to ensure the amount of light on this meter does not exceed its damage threshold. Compared to the high-power meter, the low-power meter has less drift and responds to changes in power faster. This gives quicker feedback to the aligner, allowing for easier alignment.

A Wincam was set up to monitor the beam profile. If the amplifier is significantly misaligned and mode mismatched, the beam profile will become distorted. Large misalignment will cause the beam profile to look lopsided, and smaller misalignments will cause the beam profile to have shoulders.

Once the the amplifier has been optimised for a low pump diode current, the pump diode current should be incremented in small steps of roughly 1-2A. Each time this is done, the thermal lensing will increase. Due to the non-uniform way in which the heating occurs, the input alignment needs to be re-optimised each time as does the

input mode-matching. The idea is that by going in small steps of diode current, the optimal alignment can be safely achieved. The amplifier should be operated at high diode currents, $\approx 9\text{A}$, so it is not worth spending a huge amount of time perfecting the input alignment and mode-matching at these currents. Its sufficient to be within a 1% of optimal.

Using measurements of the beam power and mode scans from the DBB, one can determine when the alignment to the amplifier is optimal. Since the beam from the amplifier will have a few percent of higher order modes due to thermal lensing, optimising the mode matching from the amplifier to the DBB is difficult. When optimising the pump diode currents, the thermal lensing changes, so the mode matching to the DBB needs to be corrected for this.

3.5 Intensity Stabilisation of a 100 W Beam

Read this! The PSL must provide intensity stabilised light to the interferometer. The intensity stabilisation of the laser is done in two stages. Before the the IMC is locked, the laser is stabilised by the inner loop to a level required to the lock the IMC. The intensity noise requirement for the PSL is found in [13] and reproduced in Figure 3.12. At low frequencies, the requirement is driven by the need for the intensity noise to be less than the radiation pressure noise induced by the differences in the arm powers. At 10 Hz, the laser must be stabilised to a level of $\sim 1 \times 10^{-9}$. Above 10 Hz, the requirement on the intensity noise is relaxed as there is passive filtering of the light by the PRC. The light is passively stabilised by the PMC to provide the required RF intensity noise.

There are two loops used for intensity stabilisation; the intensity stabilisation concept is shown in Figure 3.11. To obtain lock of the IMC, the inner loop is required. Once the IMC is locked, the outer loop is used to fulfil the intensity noise requirement for when the interferometer is in observing mode. During observation mode, the intensity noise after the IMC is a better signal to use than one from a signal in the PSL.

The light used to measure the intensity noise for the inner loop is a low power beam that is transmitted by the PMC. There is ~ 1 mW of light in this beam, so the laser cannot be stabilised beyond $\sim \frac{2 \times 10^{-8}}{\sqrt{\text{Hz}}}$ with this beam. Once the IMC is locked, a pick-off from the beam entering the interferometer is used to measure the intensity noise of the laser. This beam has enough power to reach the $\frac{1 \times 10^{-9}}{\sqrt{\text{Hz}}}$ shot noise requirement. As the inner-loop is in-air and is based on rigidly mounted optics, it is unable to remove noise associated with vibrating mounts.

3.5.1 Free Running Noise of the TnT Prototype PSL

The power noise of the NPRO was measured using the DBB RIN photodiode (Figure 3.13). The TnT NPRO displays a typical noise level that would be expected for this type of laser [19]. This was recorded via CDS, so the measurements near 10 kHz are affected by the anti-aliasing filter. The noise between 100Hz and 1kHz has features associated with the motion of the table and the mirror mounts.

The power noise of Amp II and Amp I were measured using the DBB photodiode, however the noise was measured with a spectrum analyser rather than CDS, allowing for measurements at higher frequencies (Figure 3.13). As the amplifiers

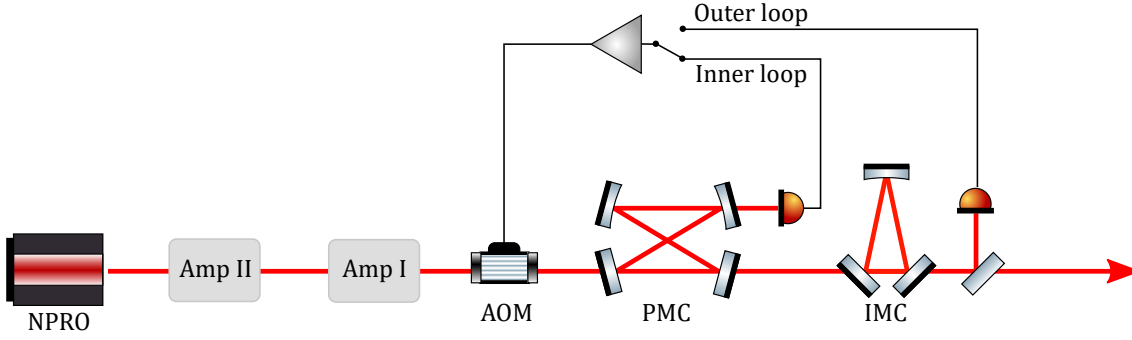


Figure 3.11: The intensity stabilisation concept involves two loops. The inner loop is responsible for stabilising the laser so that the IMC can lock. The outer loop is used to stabilise the laser for observation mode. The two loops sense the laser intensity noise at different points. The inner loop uses a low power beam transmitted by the PMC. The outer loop uses a pick-off from the main beam. An AOM will be used to actuate on the laser's power.

are pumped with low noise diodes and are operating in their saturated regimes the RIN is expected to decrease after each amplifier. When an amplifier is saturated, the amplifier will not behave linearly and instead will add a fixed amount of input power regardless of its seed power. As, expected, at frequencies above 10 kHz the RIN after the second amplifier is less than the RIN after the first amplifier. However, the noise of the amplifiers is covered in a forest of lines that are spaced 4Hz, starting from 4Hz and continuing up to high frequency. These lines were found to originate in the Beckhoff slow control computer which is used as an interface for the amplifiers. To remove these lines, this control computer needs better electrical isolation between its digital and analogue sections. There are also peaks at 20Hz, which can be attributed to the air conditioning.

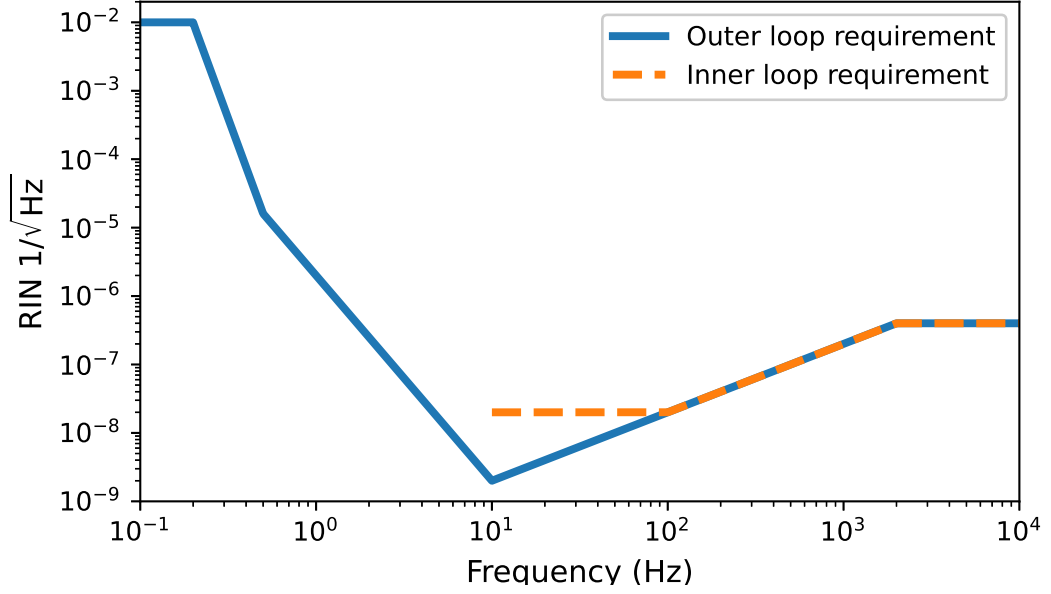


Figure 3.12: A reproduction of the intensity noise requirement from [13]. The outer loop requirement is shown in blue and the inner loop is shown by the orange dashed line. At frequencies below 10 Hz, the outer loop requirement is determined by the radiation pressure noise that is expected to arise due to imbalances in the arm power. The requirement is most stringent at 10 Hz. Due to the PRC cavity pole, the requirement relaxes above 10 Hz. The inner loop requirement is less extreme than the outer loop as it is only used to stabilise the laser in the initial locking sequence and not during an observation run.

The noise of the laser after the beam is transmitted through the PMC was measured (Figure 3.13). The PMC introduces length to intensity noise coupling via the offset in the PDH loop. This offset changes due to changing levels of RF coupling in the PMC servo electronics. As there was constant activity in the lab, this pick up changed - even moving a wire can cause a different amount of RF coupling. The coupling from length noise of the PMC to intensity noise as much as possible.

3.5.2 Power Control Actuators

The actuator that will be used to provide intensity stabilisation will be an AOM. The use of a current shunt integrated into one of the pump diodes for Amp I was investigated as this has the potential to reduce the number of transmissive high power optics. To use the current shunt, the servo electronics should be redesigned to accommodate for the difference in the actuation strength of the current shunt.

3.5.2.1 Acousto-Optic Modulator

An AOM uses acoustic waves to scatter light into higher order modes. The amount of light that is Bragg scattered depends on the intensity of the the acoustic wave. The AOM used is a Crystal Technology 3080-199. To control the intensity of the acoustic wave an RF driver, gooch and housego r21080-2am-rfx, was used. This driver produces an 80 MHz signal whose power can be modulated. By modulating the power of this signal, the power in the zeroth order beam can be controlled. The diffraction efficiency of this AOM was found to be 74%. This means that the AOM can be used for coarse and fine power adjustment. In order to modulate the intensity of a laser beam, some of it must always be scattered into the first order beam. In

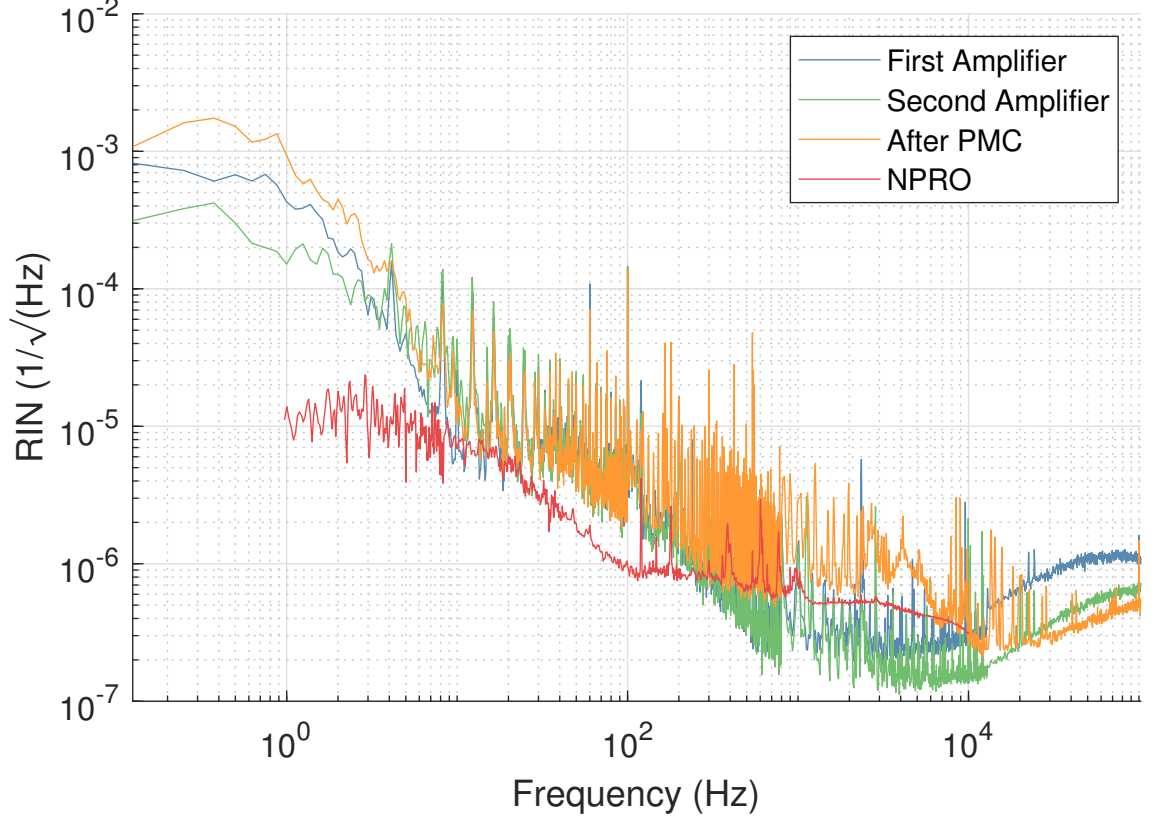


Figure 3.13: The free running noise of the NPRO and each amplifier is shown. The noise after the PMC was also measured. There is a forest of 4 Hz lines due to the Beckhoff control computer in the amplifier and PMC noise curves. Due to the non linear nature of the saturated amplifiers, at high frequencies the output beam intensity noise decreases relative to the input beam. The PMC couples length noise to intensity noise between 100 Hz and 10 kHz. This excess can be tuned away in the PDH servo.

order to operate linearly, the DC level of the light that is dumped should be about 10W.

The transfer function, in relative intensity units, between the amplifier power control of the AOM and the intensity of the zeroth order beam as measured by the ISS photodiode is shown in figure 3.15. The schematic for this measurement is shown in Figure 3.14. The signal needs to be linearised before going to the RF amplifier. This transfer function is flat from 1Hz to 100kHz, and above 10kHz there is some phase lag.

3.5.2.2 Current Shunt

Modulating the current through a laser diode will modulate the intensity of the laser. This is a standard way of implementing intensity control at the 10W level [24–28]. Using a current shunt as an intensity actuator has been demonstrated in a similar system at the 100W level [12].

The pump diodes current for the laser amplifier heads control the amount of power produced by the amplifier. By modulating the current through them, the power of the laser can be controlled. The current shunt diverts current from the pump diode by pulling it through resistor. This is controlled by a power transistor.

The current shunt technique allows for one to have fewer transmissive high power optics in the beam path, which potentially can reduce noise and thermal distortions of the beam. It requires a smaller DC offset compared to an AOM in order for it to work.

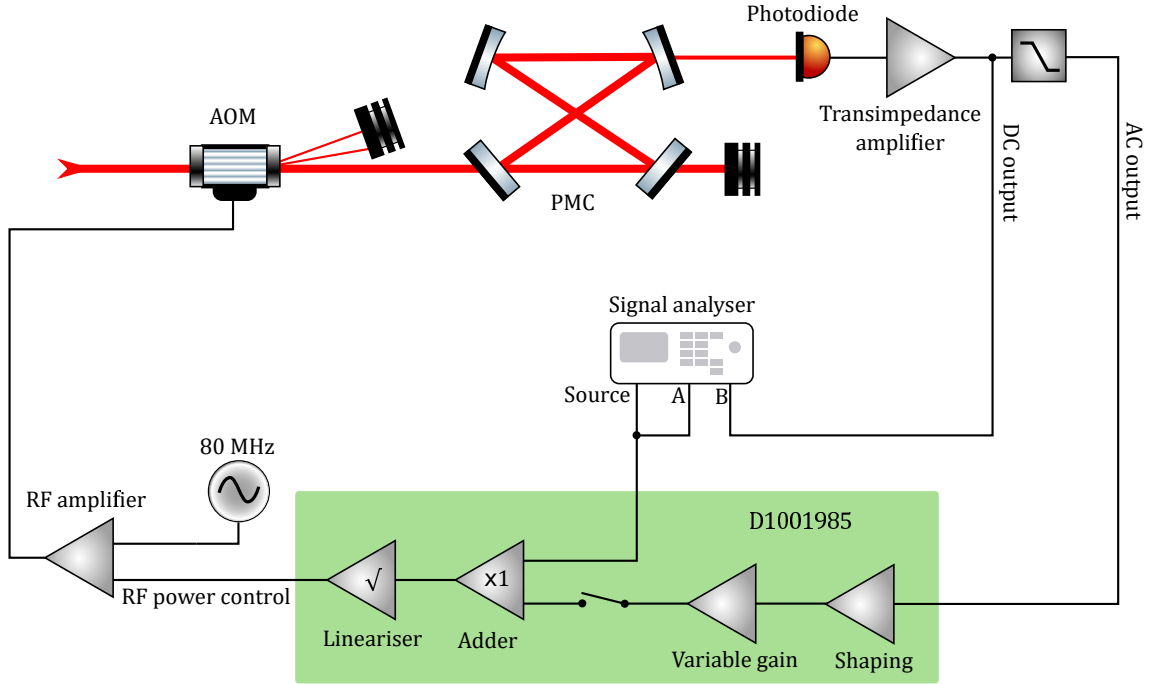


Figure 3.14: A sketch of how the AOM transfer function was measured. The components of the intensity stabilisation servo has also been shown. The PMC was locked and a signal analyser was used to perform a swept sine measurement. An IC (AD734) that performs a square root operation to an input signal was used to linearise the AOM's affect on beam power to an input signal. The transfer function was measured between the input to the lineariser and the photodiode used for the intensity stabilisation. The servo electronics are detailed in D1001985 and D1001998.

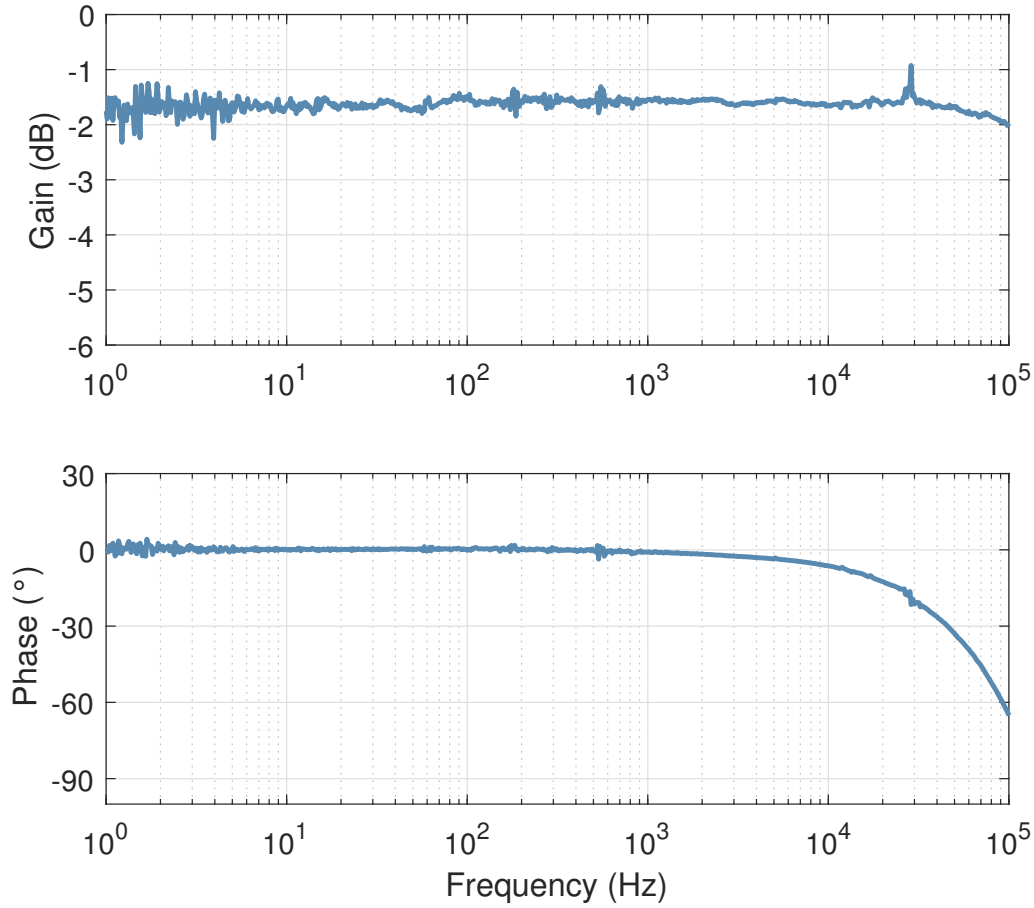


Figure 3.15: Using the layout shown in figure 3.14, the transfer function between the input to the linearisation stage before the AOM to the light power change on the ISS photodiode was measured. The response was normalised by dividing by the DC power on the photodiode. The transfer function is flat over the region where there open loop gain needs to be greater than one (~ 10 kHz), There is a small amount of phase lag at higher frequencies.

The transfer function between the diode current input and the ISS photodiode in transmission of the PMC was measured for a variety of different pump diode operating currents. A sketch of this is shown in figure 3.16. The transfer function was found to have significant amounts of variation in gain and phase between 1Hz-100kHz. It was found that the behaviour of the actuator at low frequencies strongly depended on the DC pump diode current. However, above 10kHz the transfer function was less dependant on the DC pump diode current. This is good, as the unity gain point of the intensity stabilisation servo will be between 10-100kHz, so the servo is more robust to changes in pump diode current. Figure 3.17 shows the transfer function, in relative intensity units, for the current shunt actuator with a variety of different current shunt offsets. The AOM is a much stronger actuator than the current shunt, by roughly 30 dB. This can be compensated for by changing the servo electronics.

3.5.2.3 Reasoning for Using the AOM Rather than the Current Shunt

While the current shunt offers some advantages over the AOM, it cannot reduce the power at the many tens of Watts level which is a functionality that LIGO would likely need. LIGO has never operated at full power due to point absorbers in the test masses. To reduce the power by many tens of Watts the pump diode DC current would have to be reduced. This would affect the noise of the laser and the open loop transfer function of the servo. These amplifiers work best in their saturated regimes and the RIN may be increased if the pump diode currents are decreased significantly. The transfer function of the current shunt heavily depends on the pump diode DC current at 10 Hz, this is where the requirement of the intensity stabilisation is most stringent. Alternatively, high power attenuation optics would be required, which means putting transmissive optics back in the beam bath and thus undoing the

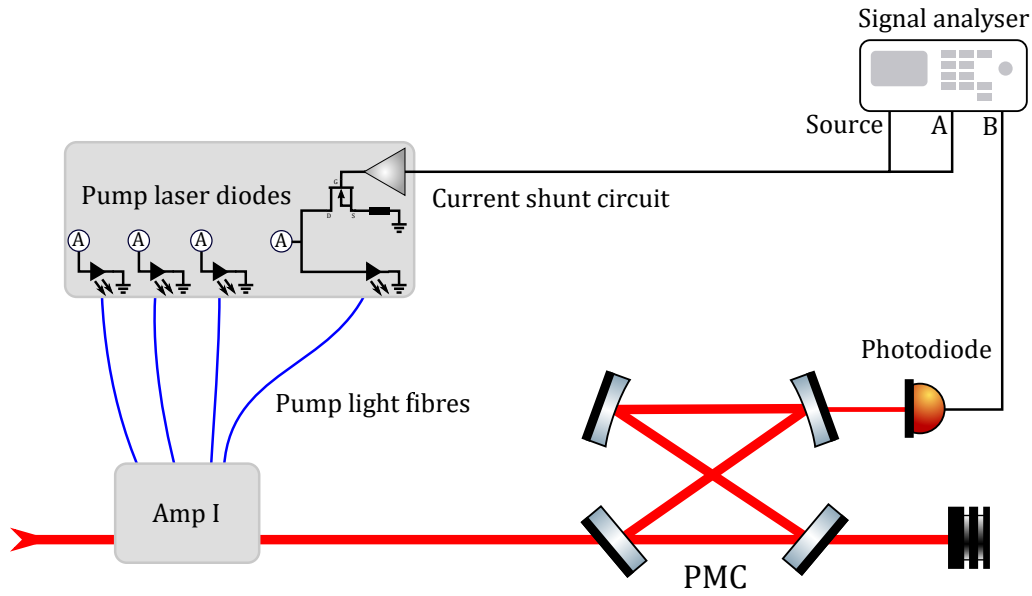


Figure 3.16: A sketch of how the current shunt transfer function was measured. The PMC was locked and a signal analyser was used to perform a swept sine measurement. The current of one of the pump diodes in the second amplifier was modulated by applying a signal to a current shunt. The transfer function was measured between the input to the current shunt electronics and the photodiode used for the intensity stabilisation.

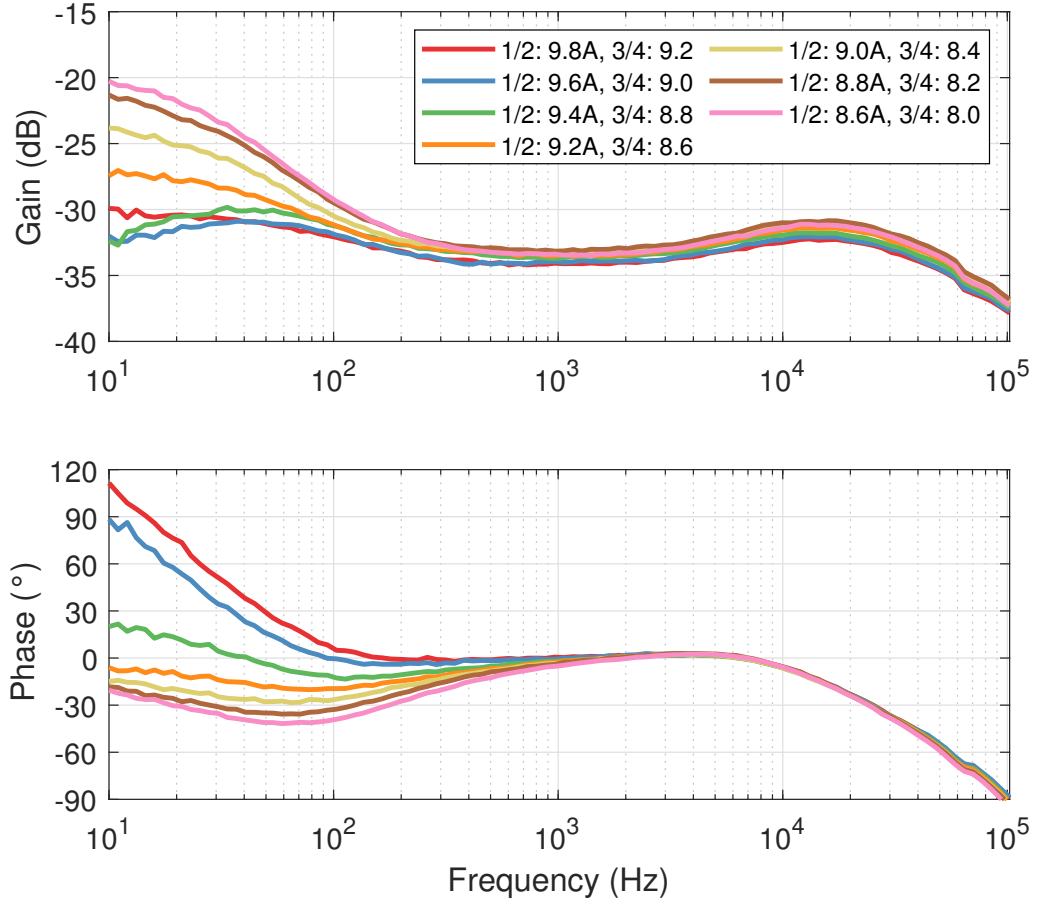


Figure 3.17: Using the layout shown in Figure 3.16, the transfer function between the current shunt input electronics and the intensity stabilisation photodiode was measured. The transfer function was normalised by dividing by the DC power on the photodiode. At 10 Hz, where the noise requirement is hardest to meet, the gain of transfer function changes significantly depending on the pump diode DC current. The phase at this frequency does not matter, as it is not near the unity gain point of the servo.

plus points gained by using a current shunt. When the LIGO detectors are able to operate at full power, then the current shunt actuator would be worth revisiting, however at present LIGO operates with between 25-50W of input power from the PSL.

3.5.3 Open Loop Transfer Function of the Intensity Stabilisation Servo

The free running noise of the laser transmitted by the PMC is shown in figure 3.13 and the requirement is shown in figure 3.12. Based on the noise and the requirement, the unity gain point of this servo must be above 10 kHz. At 10 Hz, the noise requirement is $2 \times 10^{-8} 1/\sqrt{\text{Hz}}$ and the noise is $2 \times 10^{-5} 1/\sqrt{\text{Hz}}$, so there needs to be at least 60 dB of open loop gain at this frequency.

A sketch of the servo is shown in Figure 3.18. The servo features two circuits that add shaping; these are found built into the photodiode circuit and within the rack electronics. The inner loop ISS features two photodiodes, one for in loop measurements and one for out of loop measurements. The rack electronics allow for the servo to be controlled via CDS. This gives the user the ability to turn the servo on/off, to vary the overall gain and activate integrators.

To measure the open loop transfer function, a swept sine is applied at a times one adding stage of the servo. The signal at the other input to this stage was measured and the signal immediately after this stage was measured. The ratio of these two gives the open loop transfer function of the servo. For this measurement, the gain in CDS was set to 20 dB. Once the intensity stabilisation was activated,

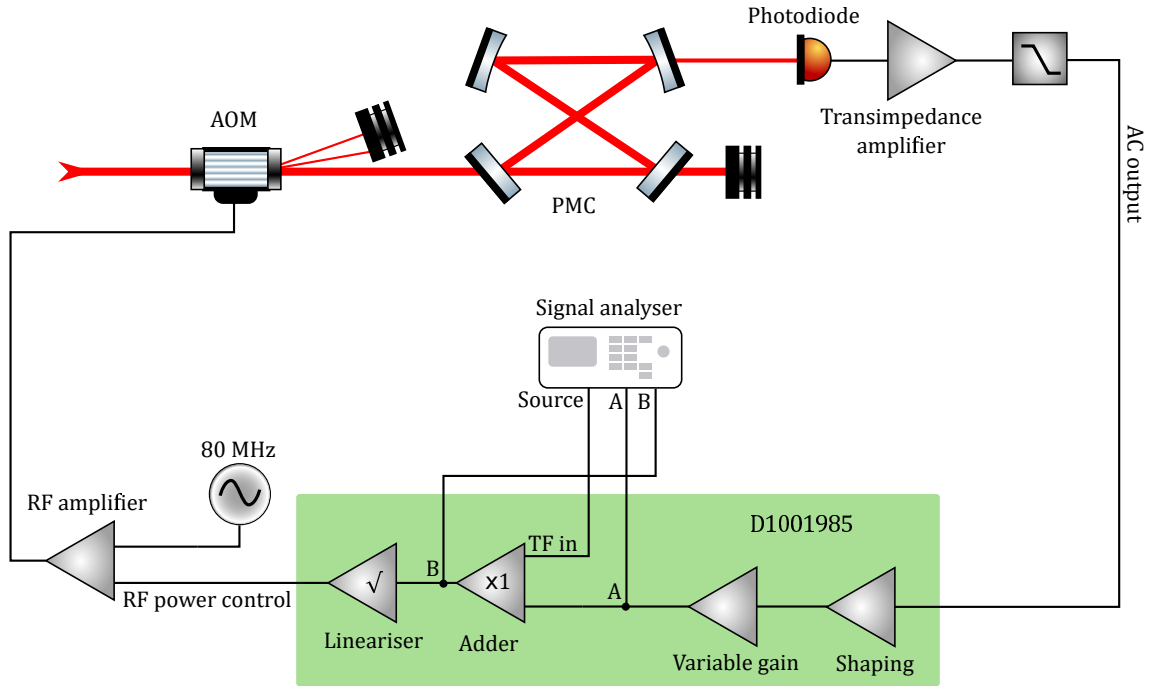


Figure 3.18: The open loop transfer function of the servo was measured by injecting a signal at a times one adding stage. The points immediately before and after this stage were measured, and the ratio of these two stages gives the open loop transfer function of the servo.

the integrators were engaged to increase the open loop gain at low frequencies. The light transmitted through the PMC was 100 W. The open loop transfer function is shown in figure 3.19. The unity gain point is at 48 kHz with $\sim 70^\circ$ of phase margin. This servo satisfies the requirements.

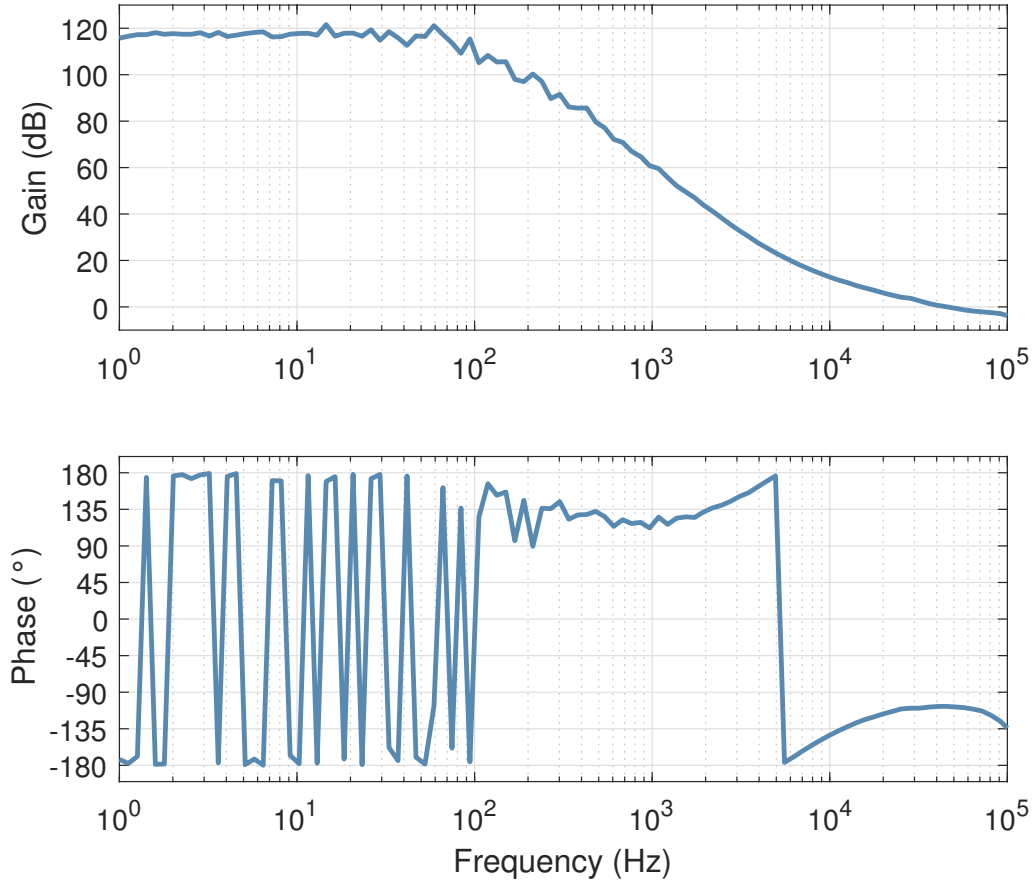


Figure 3.19: The open loop transfer function was measured using the layout shown in Figure 3.18. The unity gain point is at 48 kHz with 70° of phase margin. The shaping of the servo is $1/f$ at the unity gain point, but to achieve the higher gain required at low frequency, the servo has a $1/f^2$ slope below the unity gain point. The measurement saturated below 100 Hz.

3.5.4 Measurement of the Intensity Stabilisation of a 100 W Beam to the Inner Loop Requirement

The PMC was set to transmit 100 W of Intensity stabilised light. The stabilised laser noise is shown in Figure 3.20. A noise budget was created using the measured noise in the loop and the shot noise level of the light used to stabilise the laser. This agrees with the out of loop noise measurement above 10Hz. Below 10Hz, there is excess noise in the out of loop measurement compared to the noise budget. Possible sources of this noise could be scattered light or dust falling through the ISS beam. Near the unity gain point, due to the open loop transfer function phase and gain, there is a small increase in the stabilised noise compared to the free running noise, however this is out of the requirement frequency band.

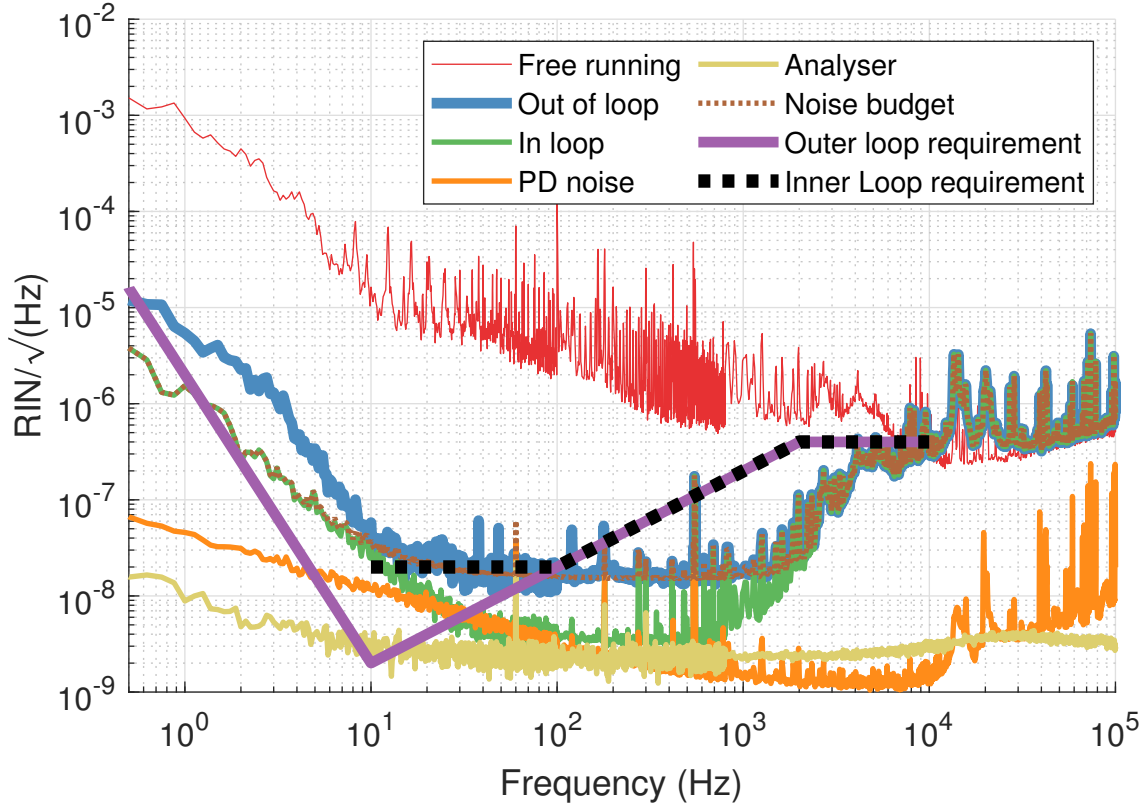


Figure 3.20: The Free running noise of the laser is shown in red. The noise of the light transmitted by the PMC is shown in blue; this meets the requirement that is shown by the dashed black line. The noise level is determined by the in loop noise, shown in green, the shot noise and other technical noise sources. The sum of these is shown by the dotted brown line. Below 100 H the noise is shot noise limited. There is a discrepancy between the noise budget and the measured noise. This is likely due to dust falling through the beam or scattered light. Near the unity gain point of the servo, the noise is enhanced by the servo due to gain peaking.

3.6 Pointing Noise of the Beam Entering the PMC

The pointing noise witnessed at LHO is one of the reasons why single-pass amplifiers were chosen as the replacement of the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. These new amplifiers are cooled in such a way that the jitter on the PSL beam is reduced. Pointing is passively stabilised by the PSL via the PMC. The pointing noise requirement is based on the transfer function between the HG10 mode between the input of the interferometer and the dark port [29].

The pointing noise was measured using the DBB. The pointing noise is shown in Figure ???. To measure pointing noise, the laser was locked the the DBB cavity. The reflected light from the cavity is made up of the the fundamental mode of the cavity (some of this mode leaks out) and the pointing modes which are excited by the laser. The beat between these two fields can be used to determine the pointing of the beam; this is known as differential wavefront sensing [30]. The electronic noise floor is at $1 \times 10^{-7}/\sqrt{\text{Hz}}$. The pointing is expressed in units of $p = \delta x/w_0 + i\delta\alpha/\Theta$, where Θ is half the divergence angle of the beam, α is the angle of the beam, x is the beams position and w_0 is the beams waist.

The pointing noise measurement shows that this layout is close to fulfilling requirement. More time could be used to improve the water cooling, as this can introduce jitter. The noise at low below 100 Hz may be coming from the environmental control in the TnT lab. The ligo PSL table is different, so this data may not reflect how quiet this layout will be in the main lab.

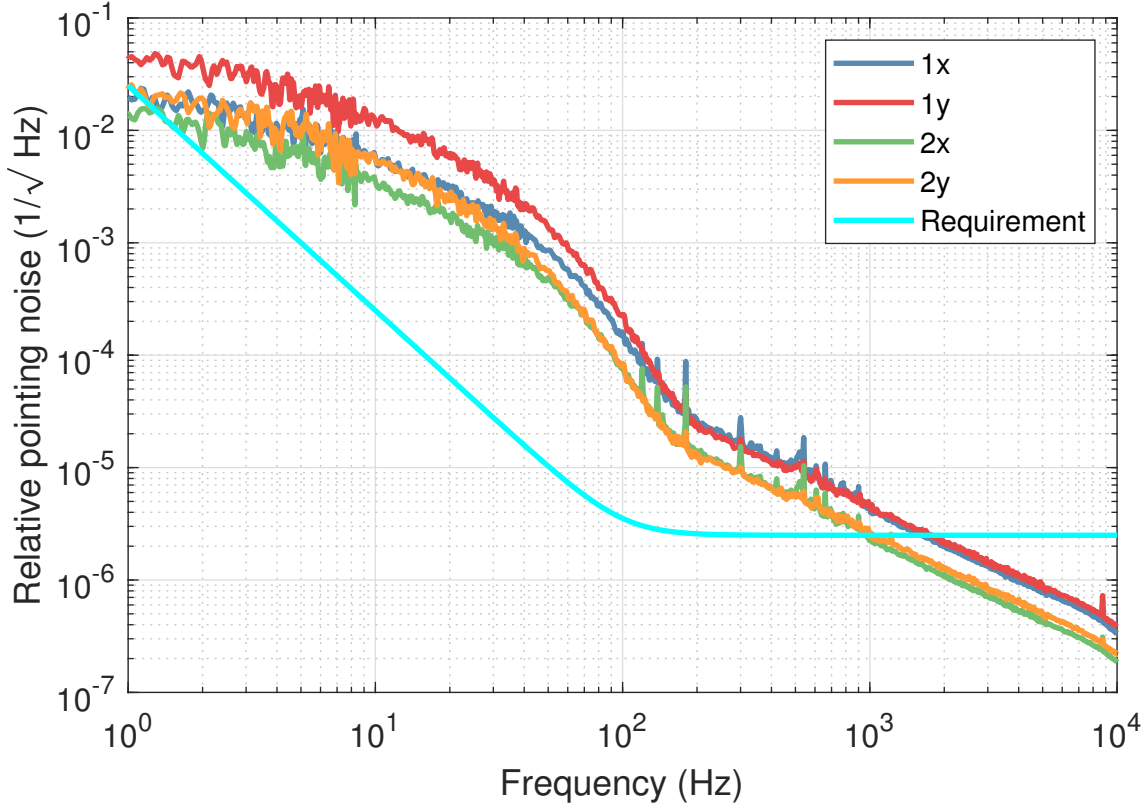


Figure 3.21: The four relative pointing noise degrees of freedom for the the beam into the PMC was measured with the DBB. The requirement [29] is shown in light blue. The pointing noise is above the requirement, however the TnT lab has a higher amount of environmental noise, this pointing noise will be less in the PSL. Measurements below 100 Hz may be affected by scattering.

3.7 Free Running Frequency Noise of the Prototype Laser System

LIGO has a multi-stage frequency stabilisation system. Ultimately, LIGO uses the arm cavity as a frequency reference since $\delta f/f = \delta l/l$, however the PSL must provide a suitably low frequency noise to the interferometer so that it can lock the first elements of the interferometer. The PSL uses an ultra low expansion fused silica reference cavity in a Pound-Drever-Hall (PDH) locking scheme to provide this frequency stabilisation [31].

The PMC can be used to measure the frequency noise of the PSL beam. The beam's frequency will be determined by the length noise of the PMC and the frequency noise of the NPRO. The signals used to control the PMC can be converted into a measurement of the systems frequency noise. The control signal which is sent to the PZT in the PMC can give the frequency noise of the system where there is sufficient open loop gain. Around the unity gain point, the servos response needs to be factored in. Alternatively, the error signal and the open loop gain can be used to measure the frequency noise of the system.

The NPRO is locked to the PMC by controlling the length of the PMC. The length of the PMC is controlled with two actuators: a PZT attached to one its mirrors and a heater which can be used to control the thermal expansion of the PMC. The PZT provides the fast control, whereas the heater controls the length of the cavity on time-scales of minutes. The Pound-Drever-Hall technique is used to sense the length of the cavity. RF frequency side-bands are applied to the laser via an EOM. The light

that is reflected from the PMC is demodulated at the same frequency as the RF side bands. When the cavity is near resonance the demodulated signal is proportional to the difference between the laser frequency and the frequency corresponding to the cavity length. A sketch of the PMC length control servo is shown in figure 3.22.

To see how the error signal used to lock the cavity can give the frequency noise, consider figure 3.23. The difference between the laser frequency and the cavity frequency is obtained via the PDH technique. The PZT signal, c , is then obtained via the transfer function of the optics, i.e. the frequency to voltage transfer function of the PDH lock, and the electronics transfer function. Together this transfer function is G . First, define the error signal $c = a - b$. Next, express the control signal in terms of the error signal, $cG = d$. Expressing the signal b in terms of d allows for signal a to be written in terms of the error signal c ,

$$a = c(1 + GH). \quad (3.1)$$

Equation 3.1 shows that to measure the frequency noise of the laser, a , the error signal should be multiplied by the open loop transfer function, GH .

3.7.1 Modelling the Open Loop Transfer Function for Locking the PMC

As the cavity is short compared to the frequencies of interest, the transfer function that converts a difference in NPRO and PMC frequency to a voltage is a flat gain factor. A measurement of this can be obtained by measuring the slope of the PDH signal as the frequency of the cavity passes through a resonance. By driving the

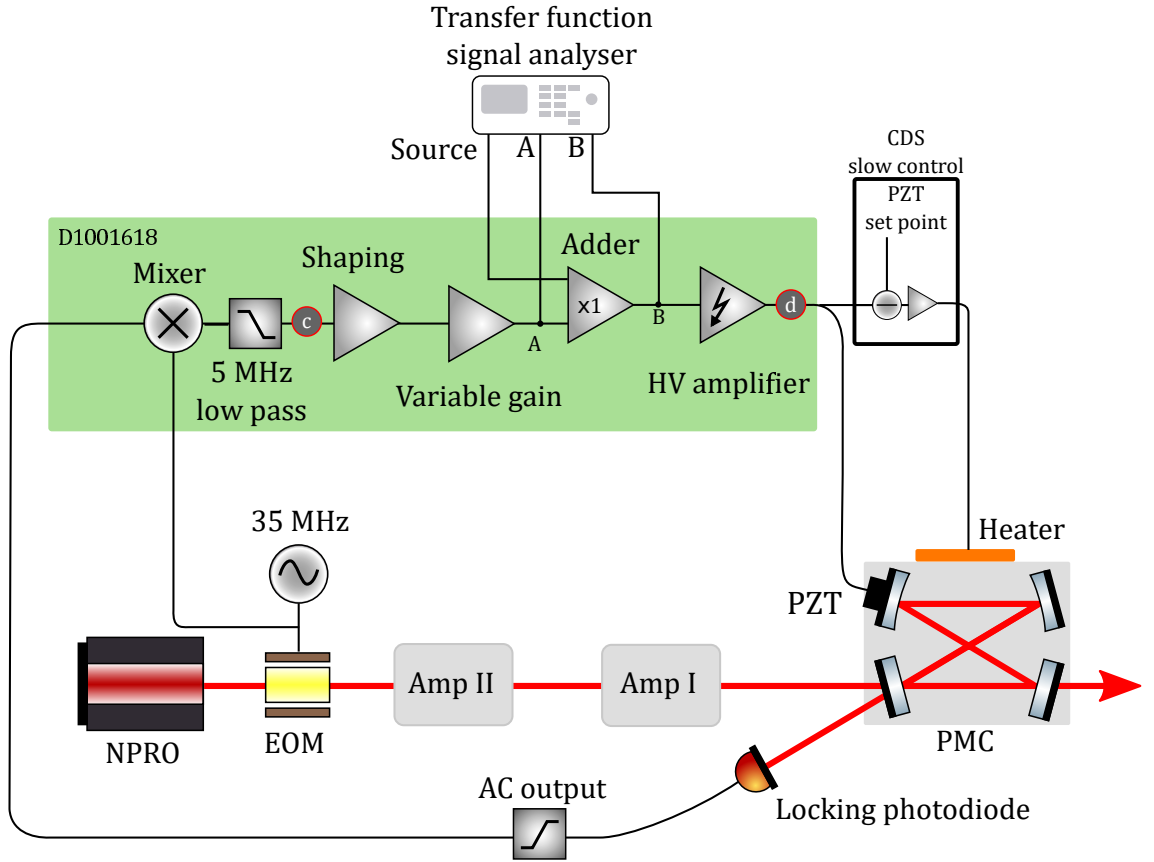


Figure 3.22: A sketch of the frequency stabilisation servo. An EOM creates 35 MHz sidebands which are used to produce a PDH locking signal. A PZT controls the length of the cavity at high frequencies. A heater and a digital loop is used to provide slow control that corrects for frequency drifts associated with change in the NPRO temperature. This is required as otherwise the PZT would run out of range. The open loop transfer function was measured using signals at A and B. The frequency noise was measured using the error point and control signals marked by c and d , with this notation corresponding with the notation in Figure 3.23 and Equation 3.1.

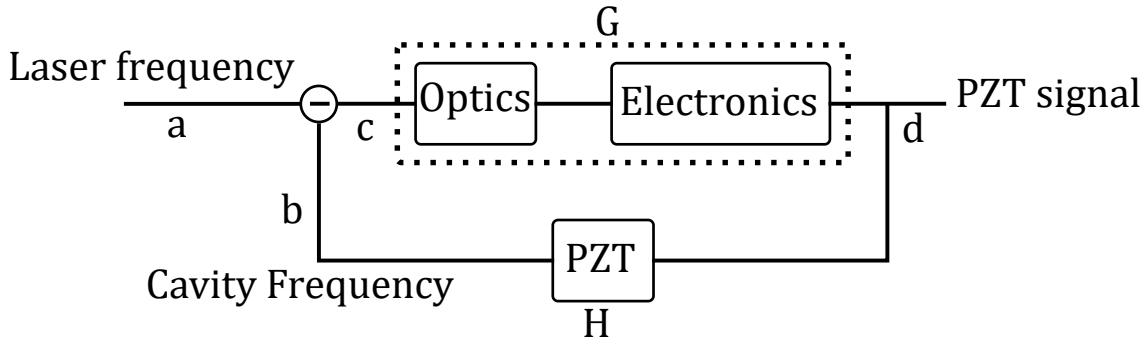


Figure 3.23: The servo can be considered as being made of three parts: the optics, the shaping electronics and the PZT. The optics block is obtained after the photodiode signal is demodulated. When at resonance and below the cavity pole, this is a flat transfer function that is proportional to the difference in frequency. Since this cavity pole is in the MHz region and this measurement will only go to the kHz region, the cavity pole can be ignored. The electronics transfer function includes everything after the demodulator up to and including the final high power OP amp. The PZT transfer function converts a voltage to a cavity frequency. The PZT acts as a low pass filter due to it having capacitance.

PZT with a ramp and simultaneously measuring the signal before and after it is demodulated, the optics transfer function can be obtained. Figure 3.24 shows this measurement. The time axis can be converted to frequency as the full-width-half-maximum (FWHM) of the reflected light is determined by the optical properties of the PMC. The FWHM for the PMC is 1.19 MHz. The slope of the demodulated signal as the cavity passes through the resonance can be used to give the conversion between difference in frequency of the cavity and laser to voltage after the demodulation electronics. This factor is 0.1364 MHz/V.

To convert the PZT voltage to the cavity frequency at low frequencies, a ramp was applied to the PZT and the spacing between two resonances was measured. The difference in ramp voltage between the two resonances corresponds the FSR of the cavity. This is shown in figure 3.25. The FSR for the PMC is 148.529 MHz, so the PZT to frequency transfer function is 42.2870 MHz/V. The PZT forms a low pass filter, as it has capacitance; this must be included when modelling the servo. The corner frequency of the PZT is above the unity gain frequency of the PMC locking servo.

The open loop transfer function, i.e. the optics, electronics and the PZT transfer functions multiplied together, is shown in figure 3.26. The PZT was modelled as a low pass filter with a corner frequency of 1700Hz. The open loop transfer function model is shown in figure 3.26. The open loop transfer function was measured, and the model is within 2 dB of the measurement. A sketch of how the open loop transfer function was measured is shown in figure 3.22. The unity gain point of the servo was measured to be at approximately 800Hz.

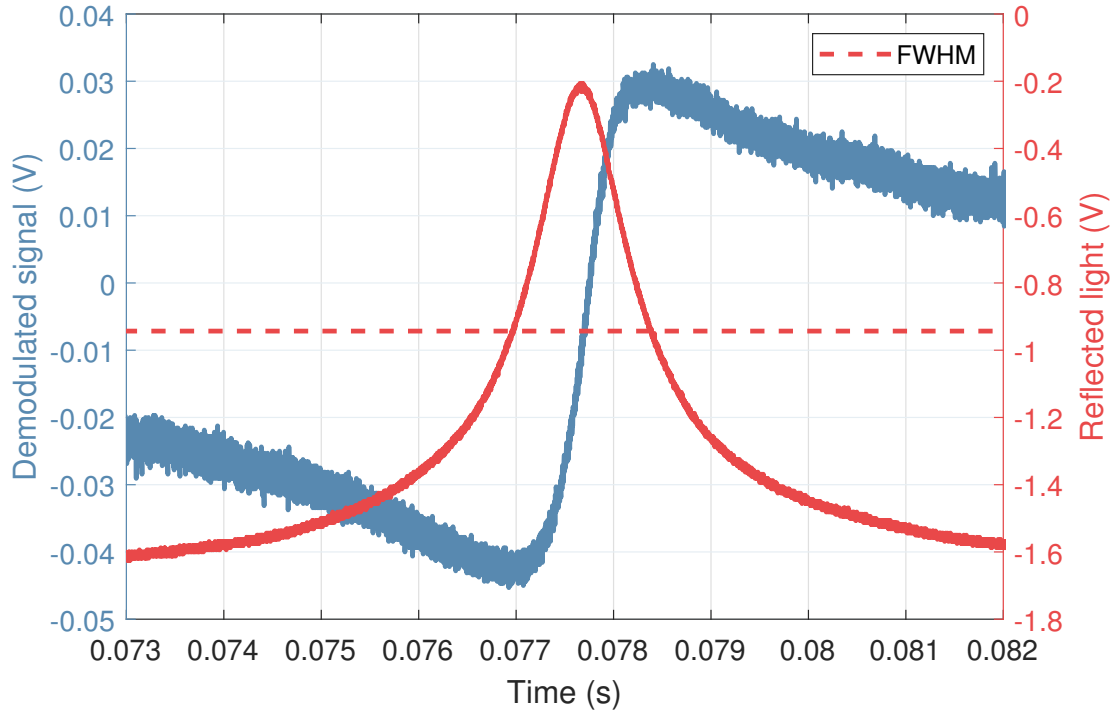


Figure 3.24: A ramp was applied to the PZT and photodiode signal was measured before and after demodulation. The demodulated photodiode signal is shown in blue. The photodiode signal is shown in red. Around the peak of the photodiode signal the slope of the demodulated signal gives the gain of the ‘optics’ transfer function. This can be expressed in units of MHz/V by converting the time axis to a frequency axis. This is done by using the FWHM of the photodiode signal as a reference frequency. The level of the FWHM is indicated by the dashed red line.

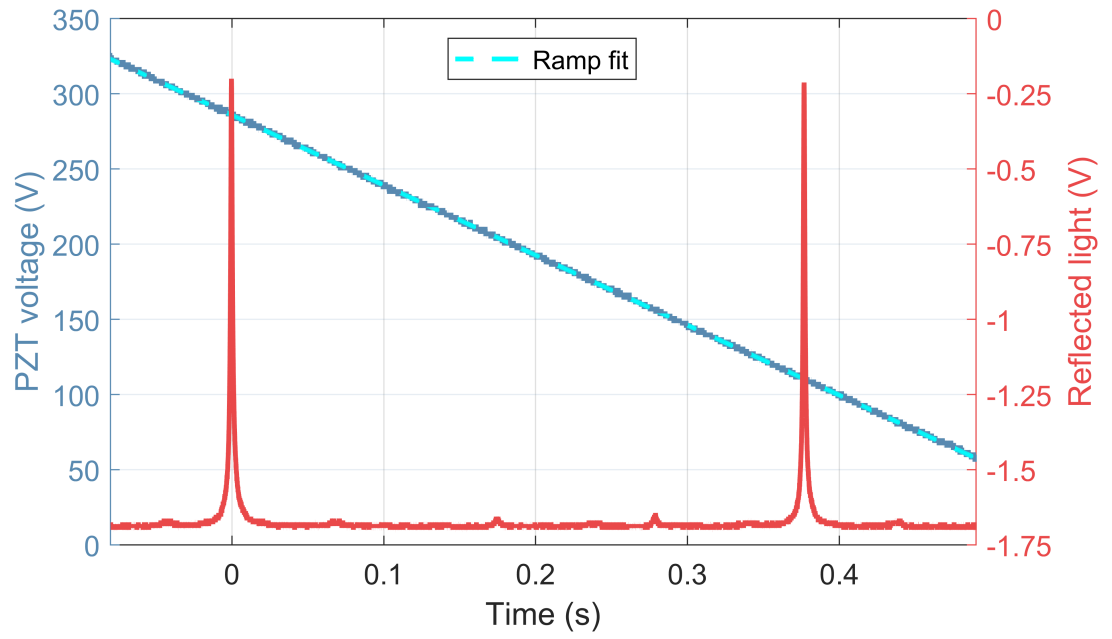


Figure 3.25: A ramp was applied to the PZT and the reflected light from the cavity was measured. The FSR of the cavity is known, so the the time in this measurement can be converted to frequency giving the PZT transfer function at low frequencies. The ramp signal is shown in blue and the reflected light is shown in red.

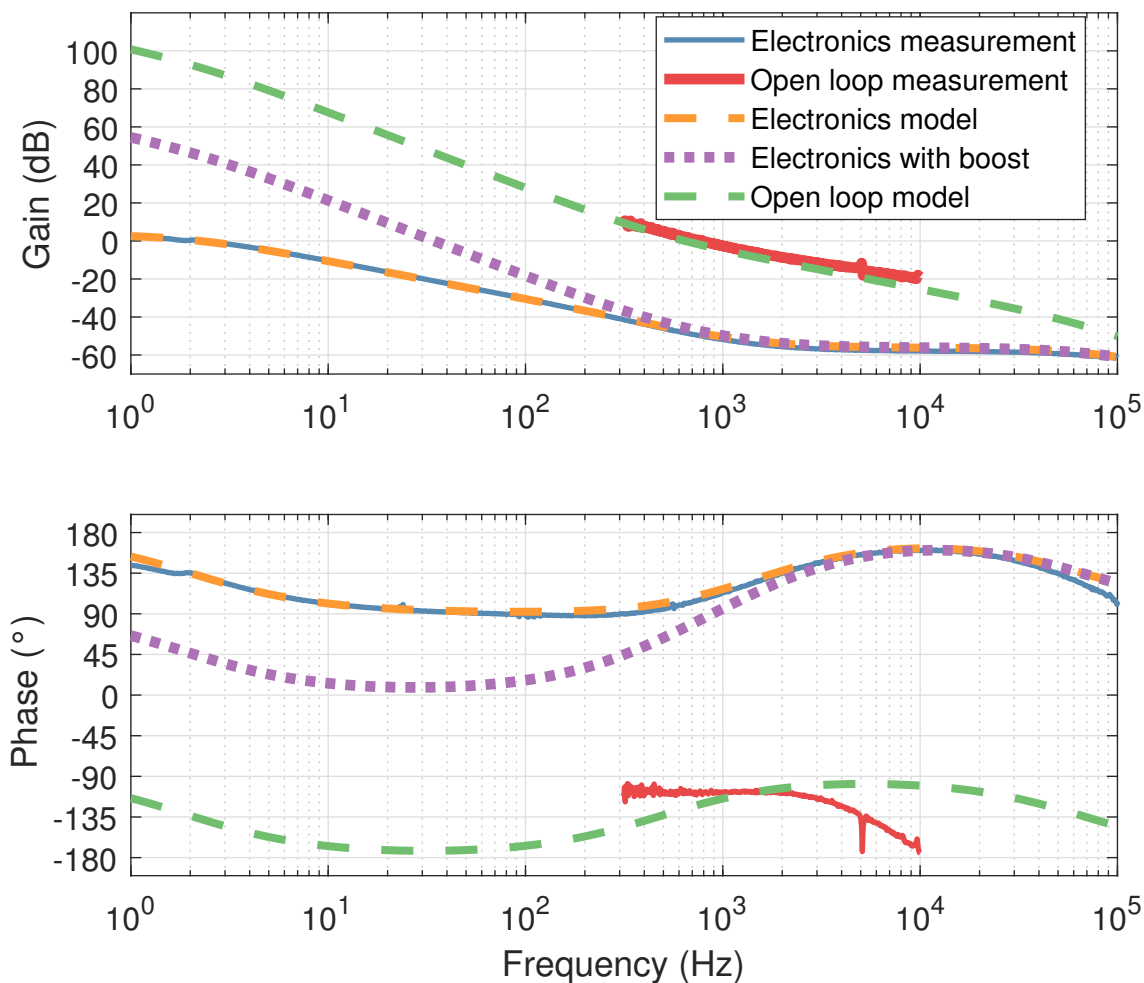


Figure 3.26: The open loop transfer function was measured using the layout shown in figure 3.22. This is shown in red. The electronics transfer function without the ‘boost’ provided by an extra integrator stage was measured and is shown in blue. The model of this is shown in orange. The boosted electronics model is shown in purple. The green line shows the expected servo gain based on the boosted electronics model and the measurements shown in Figures 3.24 and 3.25. This model is within 2 dB of the measurement. The unity gain point was measured to be at ~ 800 Hz.

3.7.2 Frequency Noise Measurement

The frequency noise is shown in figure 3.27. Below the unity gain point, as the PZT response is flat, the PZT signal can be converted into frequency noise. The mixer signal and the open loop transfer function $\epsilon(1+GH)$ gives the frequency noise. These two measurements are within a factor of 2 of each other. Above the unity gain point of the servo, a signal applied to the PZT will not be suppressed by the loop, and so a signal measured at the error point will be equivalent to the frequency noise. A peak was used to check the calibration of the loop. The discrepancy between the blue and red traces in Figure ?? may be due to the optical values used for the PMC, as these were not measured. To measure these, a broadband EOM could be used to sweep the frequency of the laser. The noise of a typical NPRO is shown. The measured noise follows this closely, it is likely that with a more accurate calibration, the PZT and error signal measurements would be closer to the expected level for NRPO noise. The 2 dB difference in the measured and modelled loop gain shows that the modelling of the loop is imperfect. The peak at 1111 Hz should overlap. The red trace below the unity gain point is better calibrated since the FSR doesn't depend on the cleanliness of mirrors. between 100 Hz to 1000 Hz, there is a bulge in the frequency noise. This comes from the HEPA filter generating acoustic noise combined with non perfect offsetting in the PDH loop. below 100 Hz and above 1 kHz, the frequency noise of the laser is determined by the NPRO. Near the UGP, there is gain peaking which will enhance the noise.

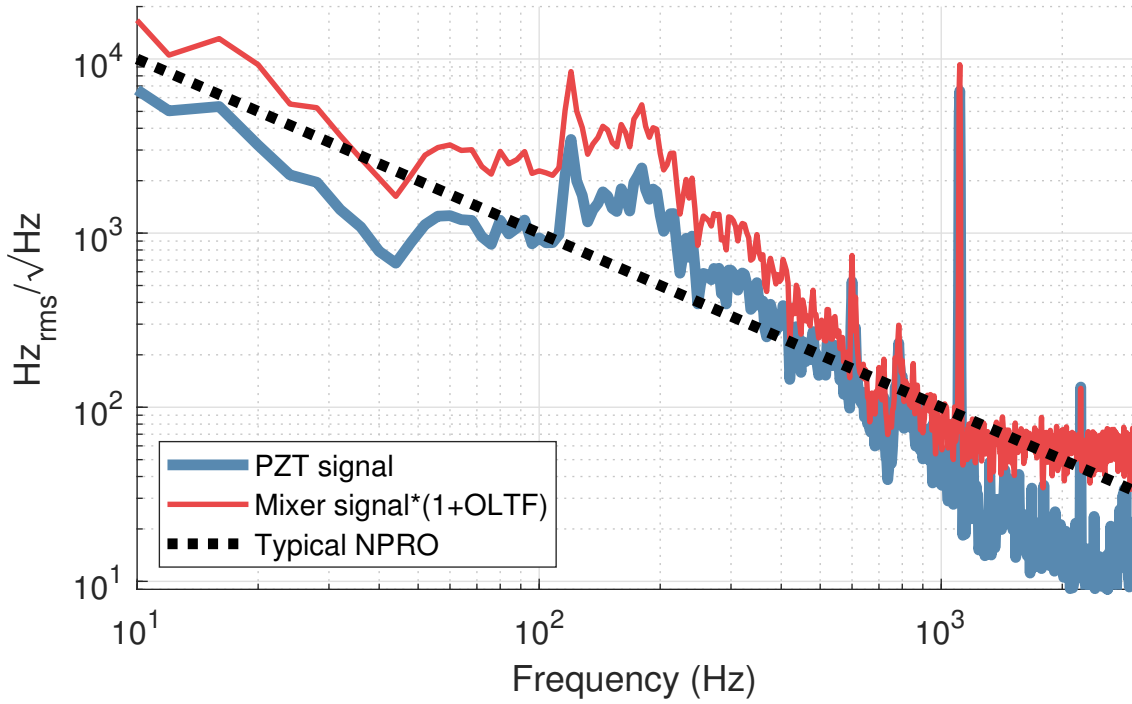


Figure 3.27: The mixer signal combined with the open loop transfer function is shown in red. This is equivalent to the frequency noise of the system. The PZT signal should agree with this measurement below the unity gain point (~ 800 Hz). A signal was applied to the PZT at 1111 Hz. This is below the corner frequency of the PZT and above the unity gain point for the servo. This should match with the mixer signal. There is a factor of two between these two measurements. This can be attributed to imperfect calibration due to nominal parameters of the PMC being used. The typical noise expected from an NPRO is shown by the dashed black line. Between 100 Hz and 1000 Hz, there is a bulge in the frequency noise. This is likely due to acoustically driven length noise of the PMC as the PMC is not vacuum sealed.

3.8 Conclusion

The PSL has undergone a major redesign due to the jitter noise introduced by the cooling for the HPO. The new layout consists of an NPRO and two solid state amplifiers followed by the PMC. A prototype of this new design was constructed and tested. The set-up procedure was established and a cooling scheme was trialled.

The output power was measured over two days and was found to drop from a peak of 111 W to 107 W. This was due to the changes the lab temperature affecting the alignment. 100W of intensity stabilised light was transmitted through the PMC. Higher output power could be achieved if more pump diode current and cooling was used. Additionally, more power could be coupled to the PMC if more time is spent on optimising the mode matching between the second amplifier and the PMC.

The frequency noise of the beam transmitted by the PMC was investigated using signals in the PMC control loop. The laser is dominated by NPRO noise at frequencies below $\sim 100\text{Hz}$ and above $\sim 1\text{kHz}$. As the PMC is not shielded from acoustic vibrations, the PMC introduce noise between these two frequencies.

The jitter noise was measured and it was found to be close to the requirements. As the PSL is in a quieter environment than the lab this layout was tested in, this may not be a problem. To reduce the jitter noise further, the flow rates being reduced may help, however this reduce the output power. Active alignment could be implemented using the light that is reflected from the PMC

The free running intensity noise of the beam transmitted by the PMC was measured and the inner loop intensity stabilisation was tested. Two actuators were

considered, a current shunt or an AOM. The AOM was chosen as this offers the most flexibility in terms of loop gains and operating power. The inner loop noise requirement was met with this set-up.

Future iterations of the PSL may use fibre lasers and fibre amplifiers [9,10]. These have been demonstrated at the power levels required for LIGO. Coherent power addition is another technique that can be used to obtain high beam powers. When long term operation has been demonstrated these technologies may be a viable replacement for the current PSL design.

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